

Supplemental Online Content

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This supplemental material has been provided by the authors to give readers additional information about their work.

eTable 1: Participating Sites

Site Name	City	State	Site Principal Investigator
Blank Children's Hospital	Des Moines	Iowa	Wendy Woods-Swofford MD, MPH
Children's Hospital Colorado	Denver	Colorado	Jenna Demedis MD, MS
Children's Hospital New Orleans	New Orleans	Louisiana	Lolie Yu MD, MPH
Children's Hospital of Richmond	Richmond	Virginia	Daniel Cannone DO
Children's Hospital Los Angeles	Los Angeles	California	David Freyer DO, MS Etan Orgel MD, MS
Columbia University	New York	New York	Melissa Beauchemin PhD
Connecticut Children's	Hartford	Connecticut	Andrea Orsey MD, MSCE
Driscoll Children's Hospital	Corpus Christi	Texas	Farha Sherani MD
Kapi'olani Medical Center for Women & Children	Honolulu	Hawaii	Wade Kyono MD
Lucile Packard Children's Hospital	Palo Alto	California	Catherine Aftandilian MD Christina Baggott PhD
MD Anderson Cancer Center	Houston	Texas	Michael Roth MD
Nemours Children's Hospital, Delaware	Wilmington	Delaware	Emi Caywood MD
Nemours Children's Hospital, Florida	Orlando	Florida	Vibhuti Agarwal MD Ramamoorthy Nagasubramanian MD
Nemours Jacksonville	Jacksonville	Florida	Scott Bradfield MD
Phoenix Children's	Phoenix	Arizona	Alexandra Walsh MD
Roswell Park Oishei Children's	Buffalo	New York	Kara Kelly MD
University of Iowa	Iowa City	Iowa	David Dickens MD
UT Health San Antonio	San Antonio	Texas	Allison Grimes MD, MSCI
Vanderbilt University Medical Center	Nashville	Tennessee	Adam Esbenshade MD, MSCI
Washington University in St. Louis	St. Louis	Missouri	Allison King MD, PhD

eTable 2: Number of Care Pathway Template Statements and Examples of Statements and Local Adaptations for each SSPedi Symptom^c

SSPedi Symptom	Care Pathway Section ^a	Number of Statements	Care Pathway Template Statement: Example	Care Pathway Statement Local Adaptation: Example ^b
Feeling disappointed or sad	Prevention	2	Promote routine physical activity	Promote routine physical activity and sleep hygiene
	Assessment	1	Assess for self-harm and suicidal ideation. If present, consult urgently	Assess for self-harm and suicidal ideation. If present, inform social work and consult psychiatry for inpatients and emergency room for outpatients urgently
	Treatment	7	When an antidepressant is initiated, monitor for self-harm, suicidal ideation or hostility. If present, consult urgently	For patients in whom an antidepressant is initiated, monitor for self-harm, suicidal ideation or hostility. If present, inform social work and consult psychiatry for inpatients and emergency room for outpatients urgently
Feeling scared or worried	Prevention	1	Encourage patient to share their feelings and concerns	With the help of social services and child life services, all health care professionals should encourage patients to share their feelings and concerns
	Assessment	0		
	Treatment	2	Consult psychiatry, psychology, social work, art therapy, chaplaincy, child life services, music therapy or recreational therapy. Encourage peer support groups	Consult social work, child life services, art therapy, spiritual care services, music therapy, psychology or psychiatry based on severity and patient preference. For AYA patients, encourage peer support groups
Feeling cranky or angry	Prevention	1	Encourage patient to share their feelings and concerns	With the help of social services and child life services, all health care professionals should encourage patients to share their feelings and concerns
	Assessment	0		
	Treatment	1	Consult psychosocial resource, particularly for violent thoughts towards self and others	Engage social work, child life services, peer support groups, art therapy, chaplaincy or music therapy if mild. Consult psychology and/or psychiatry, particularly for violent thoughts towards self and others, and engage social work or child life services if moderate to severe. Also engage peer support groups, art therapy, chaplaincy or music therapy as appropriate
Problems with thinking or remembering things	Prevention	1	No standard approaches to prevention	Promote enriched learning environment emphasizing parent or caregiver interaction with child (examples: reading books or minimizing passive screen time)
	Assessment	0		
	Treatment	2	Consult neuropsychologic and rehabilitation services as indicated	Consult neurology, psychiatry or psychology as indicated. For patients >15 years old, consult AYA education or career specialist as indicated
Changes in how your body or face feels	Prevention	1	Consider consulting a mental health specialist for patients at risk of bother due to expected changes in physical appearance	Consult child life services, music therapy, art therapy, psychology, social work or psychiatry for patients at risk of bother due to expected changes in physical appearance
	Assessment	0		
	Treatment	2	Offer adaptive devices such as prostheses or wigs	Offer adaptive devices such as prostheses (orthopedic surgery) or wigs (social work)
Feeling tired	Prevention	4	For red cell transfusions, follow institutional standards	Transfuse red cells for Hgb<7 g/dL or as per individual care plan
	Assessment	0		
	Treatment	6	Promote relaxation	Promote relaxation and/or mindfulness and engage child life services, social work, art therapy, chaplaincy or music therapy as indicated
Mouth Sores	Prevention	6	For patients at high risk, consider using cryotherapy if associated chemotherapy given by short IV infusion and co-operative patient	In cooperative patients at high risk, use cryotherapy (e.g. ice chips or ice pops) during administration of chemotherapy given by short IV infusion (e.g. bleomycin, dactinomycin, 5-FU, idarubicin, melphalan, mitoxantrone)
	Assessment	1	Evaluate oral mucous membranes routinely	Evaluate oral mucous membranes daily when in hospital and routinely when in clinic
	Treatment	3	Optimize nutrition including the use of enteral feeds or parenteral nutrition. Consult clinical nutrition services	Optimize nutrition and schedule evaluation by registered dietitian. Use enteral feeds or parenteral nutrition as indicated
Headache, hurt or pain	Prevention	18	Patients and their parents should be educated and given options for oncologic procedures. For lumbar	Patients and their parents should be educated and given options for oncologic procedures. For lumbar puncture procedures, use topical anesthetic

			puncture procedures, use topical anesthetic and offer deep sedation or general anesthesia	
	Assessment	0		
	Treatment	8	For acute pain, use heat/cold, topical anesthetics, acetaminophen or opioids with an aim to optimize pain control and minimize opioid exposure	Use heat/cold, topical anesthetics, acetaminophen or opioids with an aim to optimize pain control and minimize opioid exposure and follow institutional standards
Tingly or numb hands or feet	Prevention	1	No standard approaches to prevention	Encourage physical activity
	Assessment	0		Assess for co-existing steroid-induced myopathy
	Treatment	9	Use amitriptyline, duloxetine, gabapentin or pregabalin as initial treatment for neuropathic pain	Use gabapentin as initial treatment for neuropathic pain based on severity
Throwing up or feeling like you may throw up	Prevention	16	In patients receiving highly emetogenic chemotherapy, use palonosetron if dexamethasone and (fos)aprepitant are not permitted	If dexamethasone and (fos)aprepitant not permitted, use ondansetron
	Assessment	0		
	Treatment	7	For patients with breakthrough vomiting or nausea who are receiving highly emetic chemotherapy, consider adding olanzapine If olanzapine not possible, consider methotrimeprazine or metoclopramide	For patients receiving highly emetic chemotherapy, add olanzapine or metoclopramide + diphenhydramine (to minimize dystonic reactions) If these agents cannot be used, use lorazepam, diphenhydramine, prochlorperazine or dronabinol
Feeling more or less hungry than you usually do	Prevention	1	No standard approaches to prevention	Consult dietitian at diagnosis
	Assessment	1	Assess nutritional status including growth parameters	Assess nutritional status including growth parameters and BMI
	Treatment	3	Suggest pharmacological agents not be routinely used	If dietary modification is ineffective, consult nutrition services and use cyproheptadine, dronabinol or megestrol acetate based on patient factors
Changes in taste	Prevention	1	No standard approaches to prevention	All sites adopted this statement. No adaptations were made.
	Assessment	0		
	Treatment	1	Provide dietary counseling and assistance with additional seasoning of food, avoiding unpleasant food and expanding dietary options	Provide dietary counseling and assistance with additional seasoning of food, avoiding unpleasant food and expanding dietary options and consult registered dietitian
Constipation (hard to poop)	Prevention	2	Encourage age-appropriate fluid and dietary fiber intake	All sites adopted this statement. No adaptations were made.
	Assessment	1	Evaluate stool patterns routinely	Evaluate stool patterns at every visit
	Treatment	10	If non-severe constipation or after treatment for severe constipation, use PEG 3350 + electrolytes and adjust the dose according to symptoms and response. If PEG 3350 + electrolytes is not tolerated, substitute a stimulant laxative	If non-severe constipation or after treatment for severe constipation, use MiraLAX and adjust the dose according to symptoms and response. If not effective or not tolerated, add or substitute with bisacodyl or senna (stimulant laxatives) If stools are hard, add docusate or lactulose
Diarrhea (watery, runny poop)	Prevention	2	Suggest probiotics not be used routinely for prevention of <i>Clostridioides difficile</i> infection	Do not routinely use probiotics for prevention of <i>Clostridioides difficile</i> infection
	Assessment	2	Send stool sample for microbiological investigation	If clinically indicated, send stool sample for microbiological investigation
	Treatment	11	If non-severe <i>Clostridioides difficile</i> infection, use either metronidazole or oral vancomycin for initial treatment	Use oral vancomycin for initial treatment of <i>Clostridioides difficile</i> infection

^a All symptoms are assessed using SSPedi

^b Local adaptation could include deletion of the template recommendation or addition of a new recommendation

^c The care pathway template creation and adaptation processes are fully described elsewhere.¹⁻³ In brief, clinical practice guidelines that address SSPedi symptoms were identified systematically and a multi-disciplinary team created a care pathway template for each SSPedi symptom based on the clinical practice guideline recommendations. The study team then worked with two clinicians at each intervention site to adopt, adapt or reject each of 135 statements in the 14 care pathway templates ('Headache' and 'Pain other than Headache' were combined). The draft locally adapted care pathways were finalized after distribution to wider institutional teams for feedback.

eTable 3: Distribution of Care Pathway Adaptation Decisions by Site^a

	Site 1	Site 2	Site 3	Site 4	Site 5	Site 6	Site 7	Site 8	Site 9	Site 10
n	135	135	135	135	135	135	135	135	135	135
Adaptation Choice										
Keep as is	53 (39.3)	55 (40.7)	64 (47.4)	43 (31.9)	58 (43.0)	42 (31.1)	60 (44.4)	57 (42.2)	56 (41.5)	63 (46.7)
Adapt	67 (49.6)	57 (42.2)	66 (48.9)	67 (49.6)	68 (50.4)	68 (50.4)	63 (46.7)	67 (49.6)	68 (50.4)	66 (48.9)
Reject	15 (11.1)	23 (17.1)	5 (3.7)	25 (18.5)	9 (6.7)	25 (18.5)	12 (8.9)	11 (8.2)	11 (8.2)	6 (4.4)
Adaptations										
Resource added	9 (6.7)	4 (3.0)	21 (15.6)	19 (14.1)	2 (1.5)	2 (1.5)	10 (7.4)	7 (5.2)	9 (6.7)	13 (9.6)
Resource partially removed	14 (10.4)	11 (8.1)	16 (11.9)	9 (6.7)	15 (11.1)	14 (10.4)	13 (9.6)	16 (11.9)	13 (9.6)	13 (9.6)
Resource all removed	1 (0.7)	6 (4.4)	0 (0.0)	5 (3.7)	1 (0.7)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.7)	0 (0.0)
Follow institutional standards	3 (2.2)	3 (2.2)	2 (1.5)	2 (1.5)	3 (2.2)	5 (3.7)	3 (2.2)	5 (3.7)	5 (3.7)	1 (0.7)
Removed follow institutional standards	2 (1.5)	2 (1.5)	1 (0.7)	0 (0.0)	0 (0.0)	2 (1.5)	1 (0.7)	1 (0.7)	1 (0.7)	3 (2.2)
Action based on severity	20 (14.8)	15 (11.1)	19 (14.1)	27 (20.0)	30 (22.2)	22 (16.3)	16 (11.9)	21 (15.6)	25 (18.5)	21 (15.6)
Changed conditional to strong	15 (11.1)	11 (8.1)	16 (11.9)	14 (10.4)	12 (8.9)	8 (5.9)	15 (11.1)	16 (11.9)	20 (14.8)	19 (14.1)
Substantially changed content	38 (28.1)	13 (9.6)	29 (21.5)	28 (20.7)	31 (23.0)	26 (19.3)	27 (20.0)	24 (17.8)	24 (17.8)	21 (15.6)
Medication removed	4 (3.0)	5 (3.7)	3 (2.2)	10 (7.4)	8 (5.9)	6 (4.4)	5 (3.7)	5 (3.7)	8 (5.9)	11 (8.1)
Medication added	4 (3.0)	2 (1.5)	3 (2.2)	7 (5.2)	9 (6.7)	4 (3.0)	2 (1.5)	2 (1.5)	8 (5.9)	7 (5.2)
Medication substituted with a related medication	2 (1.5)	5 (3.7)	5 (3.7)	7 (5.2)	6 (4.4)	5 (3.7)	6 (4.4)	5 (3.7)	5 (3.7)	4 (3.0)
Changed category or combined	0 (0.0)	1 (0.7)	5 (3.7)	1 (0.7)	2 (1.5)	0 (0.0)	2 (1.5)	0 (0.0)	0 (0.0)	0 (0.0)
Removed or modified hyperlink	3 (2.2)	2 (1.5)	0 (0.0)	3 (2.2)	2 (1.5)	4 (3.0)	3 (2.2)	1 (0.7)	2 (1.5)	1 (0.7)
Rejections										
Not relevant for primary team	0 (0.0)	5 (3.7)	0 (0.0)	3 (2.2)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)
Resource unavailable	1 (0.7)	3 (2.2)	0 (0.0)	0 (0.0)	0 (0.0)	3 (2.2)	0 (0.0)	0 (0.0)	1 (0.7)	0 (0.0)
Did not agree	8 (5.9)	2 (1.5)	0 (0.0)	15 (11.1)	4 (3.0)	14 (10.4)	4 (3.0)	2 (1.5)	7 (5.2)	0 (0.0)
Reason not available	6 (4.4)	13 (9.6)	5 (3.7)	7 (5.2)	5 (3.7)	8 (5.9)	8 (5.9)	9 (6.7)	3 (2.2)	6 (4.4)

^a The table shows number (%)

eTable 4: Number Off-Protocol (Symptom Screening Only) and Off-Study (Both Groups) and Reasons

	Symptom Screening n=221	Usual Care n=224
Medical Record Review Only, n (%)	4 (1.8)	11 (4.9)
Off-Protocol Reason, n (%)	217 (98.2)	NA
Refusal to continue with study intervention	5 (2.3) (2 switched medical record review only)	
Physician determination	0	
Death	0	
Completion of planned study observations	212 (97.7)	
Off-Study Reason, n (%)		
Withdrawal of consent	1 (0.5)	0
Physician determination	0	0
Death	0	0
Completion of planned study observations	220 (99.5)	224 (100)

Abbreviation: NA – not applicable

eTable 5: Number of Participants Remaining on Study and Completing SSPedi and other PROs at Baseline, Week 4 and Week 8

	Baseline	Week 4	Week 8
Symptom Screening Sites, n (%)	n=221	n=221	n=221
Remained on study	220 (99.5)	220 (99.5)	220 (99.5)
Completed SSPedi	216 (97.7)	208 (94.1)	198 (89.6)
Completed other PROs	216 (97.7)	211 (95.5)	202 (91.4)
Usual Care Sites, n (%)	n=224	n=224	n=224
Remained on study	224 (100)	224 (100)	224 (100)
Completed SSPedi	213 (95.1)	207 (92.4)	209 (93.3)
Completed other PROs	213 (95.1)	207 (92.4)	209 (93.3)

Abbreviations: PRO – patient-reported outcomes; SSPedi – Symptom Screening in Pediatrics Tool

eTable 6: Fully Adjusted Primary Analysis of the Week-8 Total SSPedi Score

	Estimate	Lower 95% CI	Upper 95% CI	P Value
(Intercept)	10.8	6.2	15.4	<.001
Intervention vs. Control Group	-3.8	-6.4	-1.2	.007
Number Cancer Patients 8-18 Years who Speak English or Spanish per Year Less than Median vs. Above Median	-0.6	-3.8	2.6	.68
Percentage of Patients with Private Insurance Less than Median vs. Above Median	2.0	-1.1	5.2	.19
Age Group in Years				
8-10	Reference			
11-14	0.2	-2.1	2.6	.84
15-18	0.7	-1.5	3.0	.54
Diagnosis Group				
Leukemia/lymphoma	0.3	-3.5	4.1	.89
Solid tumor	-1.2	-5.0	2.7	.55
Brain tumor	Reference			

Abbreviation: CI – confidence interval; SSPedi – Symptom Screening in Pediatrics Tool

eTable 7: Fully Adjusted Analysis of the Baseline Total SSPedi Score^a

	Estimate	Lower 95% CI	Upper 95% CI	P Value
(Intercept)	9.6	5.7	13.5	<.001
Intervention vs. Control Group	-1.8	-3.5	-0.1	.04
Number Cancer Patients 8-18 Years who Speak English or Spanish per Year Less than Median vs. Above Median	1.2	-0.9	3.3	.24
Percentage of Patients with Private Insurance Less than Median vs. Above Median	0.5	-1.5	2.6	.59
Age Group in Years				
8-10	Reference			
11-14	0.7	-1.6	3.0	.55
15-18	0.8	-1.4	3.1	.48
Diagnosis Group				
Leukemia/lymphoma	2.7	-0.8	6.3	.13
Solid tumor	2.8	-0.8	6.5	.12
Brain tumor	Reference			

Abbreviation: CI – confidence interval; SSPedi – Symptom Screening in Pediatrics Tool

^a As an ad-hoc analysis, we evaluated baseline total SSPedi score in the fully adjusted model. In the original statistical analytic plan, we hypothesized that the baseline total SSPedi score might reflect benefit of the cluster's treatment assignment as some of the care pathway interventions might reduce symptoms prior to enrollment. This analysis shows that baseline SSPedi score was significantly better in the intervention group, suggesting that care pathways alone improve symptom scores.

eTable 8: Absolute Risk Difference in Week-8 Severely Bothersome Symptoms

SSPedi Symptom	Absolute Risk Difference^a (95% Credible Interval)
Feeling disappointed or sad	-2.6 (-8.3, 1.8)
Feeling scared or worried	-0.5 (-4.3, 3.3)
Feeling cranky or angry	-3.0 (-7.8, 1.0)
Problems with thinking or remembering things	-3.2 (-9.1, 2.0)
Changes in how your body or face look	-8.0 (-13.2, -3.3)
Feeling tired	-10.7 (-20.5, -1.4)
Mouth sores	-3.5 (-7.0, -1.1)
Headache	0.3 (-4.1, 5.0)
Hurt or pain (other than headache)	0.2 (-4.6, 5.2)
Tingly or numb hands or feet	-2.7 (-7.2, 0.9)
Throwing up or feeling like you may throw up	-1.8 (-10.4, 7.1)
Feeling more or less hungry than you usually do	-9.1 (-17.8, -1.0)
Changes in taste	-5.7 (-14.5, 1.6)
Constipation (hard to poop)	-3.0 (-7.1, 0.1)
Diarrhea (watery, runny poop)	-0.8 (-4.9, 2.7)

Abbreviation: SSPedi – Symptom Screening in Pediatrics Tool

^a This post-hoc analysis focuses on week-8 SSPedi scores as a dichotomous variable. It shows the absolute risk difference between groups on week-8 severely bothersome individual SSPedi symptoms (SSPedi score of 3 or 4) estimated from mixed models. The unadjusted absolute risk differences are shown as percentages.

eTable 9: Primary and Secondary Patient-Reported Outcomes at Week 4 by Group^a

	Week 4	
	Symptom Screening	Usual Care
<i>Number Completing SSPedi</i>	n=208	n=207
Primary Outcome		
	Mean (SD)	Mean (SD)
Total SSPedi Score ^a	8.4 (7.5)	11.7 (8.2)
Secondary Outcomes		
	n (%)	n (%)
Severely Bothersome SSPedi Symptom Scores (Score of 3 or 4)		
Feeling disappointed or sad	9 (4.3)	12 (5.8)
Feeling scared or worried	4 (1.9)	8 (3.9)
Feeling cranky or angry	6 (2.9)	13 (6.3)
Problems with thinking or remembering things	9 (4.3)	11 (5.3)
Changes in how your body or face look	11 (5.3)	27 (13.0)
Feeling tired	33 (15.9)	54 (26.1)
Mouth sores	7 (3.4)	7 (3.4)
Headache	8 (3.8)	13 (6.3)
Hurt or pain (other than headache)	12 (5.8)	12 (5.8)
Tingly or numb hands or feet	5 (2.4)	7 (3.4)
Throwing up or feeling like you may throw up	20 (9.6)	16 (7.7)
Feeling more or less hungry than you usually do	22 (10.6)	31 (15.0)
Changes in taste	11 (5.3)	25 (12.1)
Constipation (hard to poop)	10 (4.8)	13 (6.3)
Diarrhea (watery, runny poop)	8 (3.8)	4 (1.9)
<i>Number Completing Other PROs</i>	n=211	n=207
	Mean (SD)	Mean (SD)
PROMIS Fatigue Scores ^a	50.6 (12.2)	50.5 (12.8)
<i>PedsQL 3.0 Acute Cancer Module Domains Scores^a</i>		
Pain and hurt	76.1 (25.5)	73.1 (24.6)
Nausea	74.6 (21.9)	71.2 (21.5)
Procedural anxiety	72.6 (28.9)	71.3 (26.9)
Treatment anxiety	82.9 (22.6)	81.2 (22.4)
Worry	71.4 (26.4)	68.4 (24.7)
Cognitive problems	75.7 (20.8)	73.1 (20.9)
Perceived physical appearance	77.7 (25.1)	75.7 (25.8)
Communication	79.5 (21.8)	76.9 (20.7)

Abbreviations: CI – confidence interval, SD – standard deviation, SSPedi – Symptom Screening in Pediatrics Tool; PRO – patient-reported outcome

^a Primary outcome was the week-8 self-reported total SSPedi score, which is the sum of the 15 SSPedi items' scores. It ranges from 0 (no bothersome symptoms) to 60 (worst bothersome symptoms). The recall period is "yesterday or today". Secondary outcomes were the self-reported 15 individual symptom SSPedi scores (scored 0-4), fatigue and quality of life. Fatigue was measured using PROMIS Ped Bank v2.0–Fatigue where recall period was the last seven days. Higher scores represent more fatigue. Quality of life was measured using the PedsQL 3.0 Acute Cancer Module, which produces eight domain scores on a 0-100 scale where recall period was the last seven days. Higher scores indicate better health.

eTable 10: Symptom Documentation and Interventions by Group at Week 8^c

	All Participants				SSPedi Score ≥ 1				SSPedi Score ≥ 3			
	Symptom Screening	Usual Care	Difference (95% CI)	P Value ^b	Symptom Screening	Usual Care	Difference (95% CI)	P Value ^b	Symptom Screening	Usual Care	Difference (95% CI)	P Value ^b
Symptom Documentation^a												
Feeling disappointed or sad	16/220 (7.3%)	28/224 (12.5%)	0.53 (0.28, 1.01)	.05	10/57 (17.5%)	20/93 (21.5%)	0.76 (0.3, 1.91)	.56	5/5 (100%)	5/11 (45.5%)	13 (0.58, 291.4)	.09
Feeling scared or worried	26/220 (11.8%)	40/224 (17.9%)	0.63 (0.34, 1.17)	.14	9/63 (14.3%)	25/95 (26.3%)	0.52 (0.16, 1.64)	.26	4/6 (66.7%)	5/7 (71.4%)	0.8 (0.08, 8.47)	>.99
Feeling cranky or angry	7/220 (3.2%)	9/224 (4%)	0.79 (0.29, 2.15)	.80	4/61 (6.6%)	5/104 (4.8%)	1.39 (0.36, 5.38)	.73	2/5 (40%)	2/12 (16.7%)	3.33 (0.32, 34.83)	.54
Problems with thinking or remembering things	3/220 (1.4%)	2/224 (0.9%)	1.53 (0.25, 9.27)	.68	2/75 (2.7%)	1/101 (1.0%)	2.74 (0.24, 30.79)	.58	0/7 (0%)	0/14 (0%)	1.93 (0.03, 107.5)	>.99
Changes in how your body or face look	5/220 (2.3%)	8/224 (3.6%)	0.63 (0.20, 1.95)	.58	5/74 (6.8%)	4/99 (4.0%)	1.72 (0.45, 6.64)	.50	1/4 (25%)	1/21 (4.8%)	6.67 (0.32, 137.4)	.30
Feeling tired	43/220 (19.5%)	61/224 (27.2%)	0.62 (0.39, 0.97)	.04	35/144 (24.3%)	48/177 (27.1%)	0.83 (0.5, 1.39)	.48	13/28 (46.4%)	16/52 (30.8%)	1.95 (0.76, 5.03)	.22
Mouth sores	8/220 (3.6%)	8/224 (3.6%)	1.02 (0.38, 2.76)	>.99	4/21 (19%)	3/40 (7.5%)	2.9 (0.58, 14.42)	.22	0/0	2/8 (25%)	2.6 (0.04, 170.4)	>.99
Headache	15/220 (6.8%)	19/224 (8.5%)	0.78 (0.39, 1.59)	.50	8/69 (11.6%)	12/100 (12.0%)	0.96 (0.37, 2.49)	>.99	0/10 (0%)	4/10 (40%)	0.07 (0, 1.5)	.09
Hurt or pain (other than headache)	57/220 (25.9%)	46/224 (20.5%)	1.35 (0.87, 2.11)	.18	32/73 (43.8%)	31/95 (32.6%)	1.59 (0.77, 3.29)	.21	7/11 (63.6%)	4/11 (36.4%)	3.06 (0.54, 17.4)	.39
Tingly or numb hands or feet	15/220 (6.8%)	15/224 (6.7%)	1.02 (0.48, 2.14)	.97	10/57 (17.5%)	10/70 (14.3%)	1.28 (0.49, 3.32)	.63	1/4 (25%)	3/10 (30%)	0.78 (0.06, 10.86)	>.99
Throwing up or feeling like you may throw up	58/220 (26.4%)	62/224 (27.7%)	0.9 (0.59, 1.38)	.64	39/89 (43.8%)	42/104 (40.4%)	1.11 (0.61, 2.01)	.73	12/19 (63.2%)	9/27 (33.3%)	3.43 (1.00, 11.71)	.07
Feeling more or less hungry than you usually do	22/220 (10.0%)	37/224 (16.5%)	0.54 (0.30, 0.95)	.03	12/111 (10.8%)	30/130 (23.1%)	0.38 (0.18, 0.8)	.01	1/15 (6.7%)	8/35 (22.9%)	0.24 (0.03, 2.13)	.25
Changes in taste	7/220 (3.2%)	2/224 (0.9%)	3.65 (0.75, 17.76)	.10	4/72 (5.6%)	2/99 (2.0%)	2.85 (0.51, 16.02)	.24	1/11 (9.1%)	1/25 (4%)	2.40 (0.14, 42.26)	.52
Constipation (hard to poop)	20/220 (9.1%)	15/224 (6.7%)	1.4 (0.69, 2.81)	.35	6/44 (13.6%)	10/62 (16.1%)	0.82 (0.27, 2.45)	.79	1/2 (50%)	3/9 (33.3%)	2.00 (0.09, 44.35)	>.99
Diarrhea (watery, runny poop)	6/220 (2.7%)	3/224 (1.3%)	2.07 (0.51, 8.36)	.33	5/26 (19.2%)	2/56 (3.6%)	6.43 (1.16, 35.74)	.03	1/4 (25%)	0/6 (0%)	5.57 (0.18, 176.3)	.40
Any Intervention^a												
Feeling disappointed or sad	52/220 (23.6%)	82/224 (36.6%)	0.51 (0.26, 0.99)	.05	19/57 (33.3%)	42/93 (45.2%)	0.66 (0.26, 1.67)	.38	5/5 (100%)	5/11 (45.5%)	13 (0.58, 291.4)	.09
Feeling scared or worried	54/220 (24.5%)	96/224 (42.9%)	0.40 (0.19, 0.83)	.01	20/63 (31.7%)	43/95 (45.3%)	0.57 (0.23, 1.38)	.21	4/6 (66.7%)	5/7 (71.4%)	0.8 (0.08, 8.47)	>.99
Feeling cranky or angry	48/220 (21.8%)	67/224 (29.9%)	0.65 (0.32, 1.29)	.22	16/61 (26.2%)	26/104 (25.0%)	1.10 (0.47, 2.56)	.81	4/5 (80%)	6/12 (50%)	4.00 (0.34, 47.11)	.34
Problems with thinking or remembering things	5/220 (2.3%)	5/224 (2.2%)	1.02 (0.29, 3.57)	>.99	2/75 (2.7%)	2/101 (2.0%)	1.36 (0.19, 9.85)	>.99	0/7 (0%)	0/14 (0%)	1.93 (0.03, 107.5)	>.99
Changes in how your body or face look	0/220 (0%)	1/224 (0.4%)	0.34 (0.01, 8.34)	>.99	0/74 (0%)	1/99 (1.0%)	0.44 (0.02, 10.97)	>.99	0/4 (0%)	0/21 (0%)	4.78 (0.08, 274.1)	>.99

Feeling tired	12/220 (5.5%)	15/224 (6.7%)	0.47 (0.08, 2.72)	.40	9/144 (6.2%)	10/177 (5.6%)	1.11 (0.44, 2.82)	.82	3/28 (10.7%)	3/52 (5.8%)	1.96 (0.37, 10.43)	.42
Mouth sores	28/220 (12.7%)	23/224 (10.3%)	1.24 (0.38, 4.03)	.72	6/21 (28.6%)	6/40 (15.0%)	2.27 (0.63, 8.19)	.31	0/0	1/8 (12.5%)	5.00 (0.07, 366.4)	>.99
Headache	35/220 (15.9%)	42/224 (18.8%)	0.84 (0.51, 1.37)	.48	16/69 (23.2%)	21/100 (21.0%)	1.16 (0.55, 2.42)	.70	4/10 (40%)	5/10 (50%)	0.67 (0.11, 3.92)	>.99
Hurt or pain (other than headache)	68/220 (30.9%)	70/224 (31.2%)	1.00 (0.67, 1.49)	.99	34/73 (46.6%)	36/95 (37.9%)	1.43 (0.76, 2.67)	.27	7/11 (63.6%)	4/11 (36.4%)	3.06 (0.54, 17.4)	.39
Tingly or numb hands or feet	17/220 (7.7%)	25/224 (11.2%)	0.68 (0.34, 1.40)	.30	10/57 (17.5%)	14/70 (20.0%)	1.00 (0.29, 3.44)	>.99	2/4 (50%)	4/10 (40%)	1.50 (0.15, 15.46)	>.99
Throwing up or feeling like you may throw up	119/220 (54.1%)	134/224 (59.8%)	0.78 (0.53, 1.13)	.19	58/89 (65.2%)	71/104 (68.3%)	0.87 (0.47, 1.6)	.65	12/19 (63.2%)	18/27 (66.7%)	0.86 (0.25, 2.93)	>.99
Feeling more or less hungry than you usually do	18/220 (8.2%)	20/224 (8.9%)	0.90 (0.46, 1.75)	.75	13/111 (11.7%)	16/130 (12.3%)	0.95 (0.43, 2.07)	.89	0/15 (0%)	0/35 (0%)	2.29 (0.04, 120.8)	>.99
Changes in taste	2/220 (0.9%)	0/224 (0%)	5.14 (0.25, 107.6)	.25	2/72 (2.8%)	0/99 (0%)	7.06 (0.33, 149.3)	.18	1/11 (9.1%)	0/25 (0%)	7.29 (0.27, 193.7)	.31
Constipation (hard to poop)	59/220 (26.8%)	57/224 (25.4%)	1.05 (0.56, 1.99)	.87	25/44 (56.8%)	19/62 (30.6%)	2.59 (1.11, 6.03)	.03	2/2 (100%)	5/9 (55.6%)	4.09 (0.15, 108.9)	.49
Diarrhea (watery, runny poop)	0/220 (0%)	1/224 (0.4%)	0.34 (0.01, 8.34)	>.99	0/26 (0%)	1/56 (1.8%)	0.7 (0.03, 17.72)	>.99	0/4 (0%)	0/6 (0%)	1.44 (0.02, 87.17)	>.99
Intervention Clearly for Symptom (Symptom-Specific Intervention)^a												
Feeling disappointed or sad	13/220 (5.9%)	20/224 (8.9%)	0.62 (0.30, 1.28)	.19	9/57 (15.8%)	16/93 (17.2%)	0.84 (0.34, 2.08)	.71	4/5 (80.0%)	2/11 (18.2%)	18.00 (1.24, 260.9)	.04
Feeling scared or worried	21/220 (9.5%)	33/224 (14.7%)	0.59 (0.22, 1.53)	.28	9/63 (14.3%)	20/95 (21.1%)	0.71 (0.15, 3.27)	.66	4/6 (66.7%)	3/7 (42.9%)	2.67 (0.28, 25.64)	.59
Feeling cranky or angry	3/220 (1.4%)	6/224 (2.7%)	0.50 (0.12, 2.03)	.50	1/61 (1.6%)	2/104 (1.9%)	0.85 (0.08, 9.57)	>.99	1/5 (20%)	1/12 (8.3%)	2.75 (0.14, 55.17)	.51
Problems with thinking or remembering things	1/220 (0.5%)	0/224 (0%)	3.07 (0.12, 75.73)	.50	0/75 (0%)	0/101 (0%)	1.34 (0.03, 68.53)	>.99	0/7 (0%)	0/14 (0%)	1.93 (0.03, 107.5)	>.99
Changes in how your body or face look	0/220 (0%)	1/224 (0.4%)	0.34 (0.01, 8.34)	>.99	0/74 (0%)	1/99 (1.0%)	0.44 (0.02, 10.97)	>.99	0/4 (0%)	0/21 (0%)	4.78 (0.08, 274.1)	>.99
Feeling tired	2/220 (0.9%)	5/224 (2.2%)	0.40 (0.08, 2.09)	.45	2/144 (1.4%)	3/177 (1.7%)	0.82 (0.13, 4.96)	>.99	1/28 (3.6%)	1/52 (1.9%)	1.89 (0.11, 31.4)	>.99
Mouth sores	4/220 (1.8%)	2/224 (0.9%)	2.06 (0.37, 11.34)	.45	3/21 (14.3%)	2/40 (5.0%)	3.17 (0.49, 20.65)	.33	0/0	1/8 (12.5%)	5.00 (0.07, 366.4)	>.99
Headache	4/220 (1.8%)	6/224 (2.7%)	0.67 (0.19, 2.42)	.75	2/69 (2.9%)	5/100 (5.0%)	0.57 (0.11, 3.01)	.70	1/10 (10%)	2/10 (20%)	0.44 (0.03, 5.88)	>.99
Hurt or pain (other than headache)	40/220 (18.2%)	9/224 (4.0%)	5.30 (2.50, 11.24)	<.001	25/73 (34.2%)	5/95 (5.3%)	9.17 (3.29, 25.55)	<.001	7/11 (63.6%)	1/11 (9.1%)	17.50 (1.60, 191.9)	.02
Tingly or numb hands or feet	8/220 (3.6%)	0/224 (0%)	17.96 (1.03, 313.1)	.003	6/57 (10.5%)	0/70 (0%)	17.8 (0.98, 323.1)	.007	2/4 (50.0%)	0/10 (0%)	21.00 (0.75, 589.7)	.07
Throwing up or feeling like you may throw up	40/220 (18.2%)	46/224 (20.5%)	0.84 (0.52, 1.35)	.46	26/89 (29.2%)	26/104 (25.0%)	1.17 (0.56, 2.46)	.67	7/19 (36.8%)	2/27 (7.4%)	7.29 (1.31, 40.54)	.02
Feeling more or less hungry than you usually do	10/220 (4.5%)	0/224 (0%)	22.4 (1.30, 384.6)	<.001	6/111 (5.4%)	0/130 (0%)	16.08 (0.9, 288.74)	.009	0/15 (0%)	0/35 (0%)	2.29 (0.04, 120.8)	>.99

Changes in taste	2/220 (0.9%)	0/224 (0%)	5.14 (0.25, 107.6)	.25	2/72 (2.8%)	0/99 (0%)	7.06 (0.33, 149.3)	.18	1/11 (9.1%)	0/25 (0%)	7.29 (0.27, 193.7)	.31
Constipation (hard to poop)	15/220 (6.8%)	5/224 (2.2%)	3.2 (1.14, 8.98)	.02	7/44 (15.9%)	4/62 (6.5%)	2.74 (0.75, 10.02)	.19	2/2 (100%)	1/9 (11.1%)	28.33 (0.86, 936.0)	.05
Diarrhea (watery, runny poop)	0/220 (0%)	0/224 (0%)	1.02 (0.02, 51.54)	>.99	0/26 (0%)	0/56 (0%)	2.13 (0.04, 110.4)	>.99	0/4 (0%)	0/6 (0%)	1.44 (0.02, 87.17)	>.99

Abbreviation: CI – confidence interval

^a Documentation means there was some notation in the medical record of that symptom. Two intervention types were evaluated. Any intervention regardless of attribution was abstracted from a list of potential, previously-defined interventions for specific symptoms.⁵ For example, acetaminophen was considered an intervention for pain even if it was administered for fever or reason was unknown, and a child life specialist visit was considered an intervention for sadness, anxiety and anger, regardless of visit reason. Symptom-specific interventions were those where the clinical documentation clearly noted administration was for that symptom. Neither any intervention or symptom-specific intervention were specific to a site's adapted care pathway.

^b P values were calculated using a mixed or fixed effects logistic regression model depending on the number of events (details in eMethods).

^c The denominators for this table include all participants, those with any symptoms (SSPedi score ≥ 1) and those with severely bothersome symptoms (SSPedi score ≥ 3). Consequently, the denominators change with each symptom for the symptomatic populations.

eTable 11: Whether Interventions Provided were Consistent with Adapted Care Pathways for Intervention Sites at Week 8 (N=220)

SSPedi Symptom	Number of Participants Receiving Interventions Consistent with Adapted Care Pathways (%)
Feeling disappointed or sad	49 (22.3%)
Feeling scared or worried	48 (21.8%)
Feeling cranky or angry	47 (21.4%)
Problems with thinking or remembering things	0 (0%)
Changes in how your body or face look	0 (0%)
Feeling tired	11 (5.0%)
Mouth sores	0 (0%)
Headache	33 (15.0%)
Hurt or pain (other than headache)	58 (26.4%)
Tingly or numb hands or feet	17 (7.7%)
Throwing up or feeling like you may throw up	115 (52.3%)
Feeling more or less hungry than you usually do	6 (2.7%)
Changes in taste	1 (0.5%)
Constipation (hard to poop)	58 (26.4%)
Diarrhea (watery, runny poop)	0 (0%)

Abbreviation: SSPedi – Symptom Screening in Pediatrics Tool

eTable 12: Analysis of Healthcare Encounters Using Poisson Regression and Negative Binomial Models^a

Unplanned Encounter Type	RR (95% CI or CR)	P Value Adjusted	Probability RR > 1 (%)
All Unplanned Encounters			
Poisson regression (primary approach)	1.46 (0.97, 2.19)	.07	
Negative binomial frequentist	1.48 (0.98, 2.23)	.06	
Negative binomial Bayesian	1.48 (0.89, 2.48)		93.9
Emergency Department Visits			
Poisson regression (primary approach)	1.72 (1.03, 2.87)	.04	
Negative binomial frequentist	No convergence		
Negative binomial Bayesian	1.73 (0.89, 3.43)		95.2
Unplanned Clinic Visits			
Poisson regression (primary approach)	1.01 (0.41, 2.50)	.99	
Negative binomial frequentist	1.02 (0.40, 2.58)	.96	
Negative binomial Bayesian	0.98 (0.30, 3.17)		49.5
Unplanned Hospital Admissions			
Poisson regression (primary approach)	1.40 (0.96, 2.03)	.08	
Negative binomial frequentist	1.42 (0.96, 2.09)	.08	
Negative binomial Bayesian	1.41 (0.88, 2.24)		93.3

Abbreviations: RR – rate ratio; CI – confidence interval; CR – credible region

^aThe estimated effects were similar across all three models, but with CIs increasing slightly, with the Poisson models having the narrowest CIs and the Bayesian negative binomial models having the widest CIs. The substantive conclusions were the same for all models.

eFigure1: Sensitivity Analyses and Approach to Missing Data

The full statistical analytic plan was uploaded to clinicaltrials.gov before analysis began.

Primary Outcome Sensitivity Analyses

- 1. **Unadjusted Primary Outcome Model:** Model with only stratification variables of number of cancer patients and percentage of patients with private insurance resulted in mean difference -3.7, 95% CI -6.3, -1.1; P=.008.
- 2. **Square-root-transformed Outcome:** Residuals were clearly non-normal, with a right-skewed distribution. Applying a square-root transformation led to residuals that were close to normal. Applying the primary fully-adjusted outcome model to the square-root transformed total SSPedi scores resulted in an estimated effect of -0.7, 95% CI -1.1, -0.3; P=.003.
- 3. **Restriction to Patients Receiving Chemotherapy:** To account for potential differential enrollment of surgery-only patients (who might not be cared for by oncology teams) between intervention and control sites, we also conducted a sensitivity analysis restricted to patients that received chemotherapy. The percentage of patients who received chemotherapy was 210 (95.0%) in the intervention group and 207 (92.4%) in the control group. Applying the primary fully-adjusted outcome model to this subset of patients resulted in mean difference -3.8, 95% CI -6.4, -1.2; P=.01.
- 4. **Adjustment for Baseline SSPedi:** Although the pre-specified primary analysis did not adjust for baseline SSPedi, we ran the primary fully-adjusted outcome model with the baseline total SSPedi score as an additional covariate, being cognizant that this analysis adjusted for a variable in the pathway between intervention and outcome, so would likely underestimate the treatment effect. The estimated mean difference was -3.0, 95% CI -5.2, -0.8; P=.01. Notably, there was a strong relationship (slope=0.5, 95% CI 0.4-0.6; P < .001) between baseline and week-8 total SSPedi scores.
- 5. **Addressing Missing Data:** Among the 445 participants, there were 15 who declined PROs and were enrolled as medical record review-only participants. Thus, we knew they would have missing week-8 SSPedi scores. For the other participants, SSPedi scores were recorded using SPARK while the other patient-reported outcomes were recorded using REDCap. Because intervention patients were using SPARK for symptom screening, there were some intervention patients where the site clinical research associate failed to obtain the SSPedi assessment at the allotted week-4 (n=3) or week-8 (n=4) timepoint, leading to more missing SSPedi data compared to PROMIS Fatigue or PedsQL 3.0 Acute Cancer Module scores for intervention patients. The total numbers with missing SSPedi scores at the three assessment times were as follows:

Group	N	Baseline	Week 4	Week 8
Overall	445	16 (3.6%)	30 (6.7%)	38 (8.5%)
Symptom Screening	221	5 (2.3%)	13 (5.9%)	23 (10.4%)
Usual Care	224	11 (4.9%)	17 (7.6%)	15 (6.7%)

Among all 445 participants, the overall percentage with a missing week-8 SSPedi was 8.5%, which was less than the 10% *a priori* threshold for conducting multiple imputation. When excluding medical record review-only participants, the amount of missing data was 23/430 (5.3%). However, since the worst-case performance was 10.4% missing data in the intervention group among all randomized participants, we conducted five different approaches to address missing week-8 SSPedi. The correlations between baseline and week-4 (r=0.543), baseline and week-8 (r=0.532), and week-4 and week-8 (r=0.709) total SSPedi scores were high. Imputation strategies were as follows:

- a) Multiple imputation with chained equations (using the mice package in R): Variables used in the imputation model were site (as a random effect), baseline total SSPedi score, week-4 total SSPedi score, week-8 total SSPedi score, age group, diagnosis group, and, because of their relationship with the week-8 SSPedi being missing, ethnicity, first language English or Spanish, preferred language English, family composition, days from diagnosis, treatment with chemotherapy, treatment with radiation and guardian employment status.

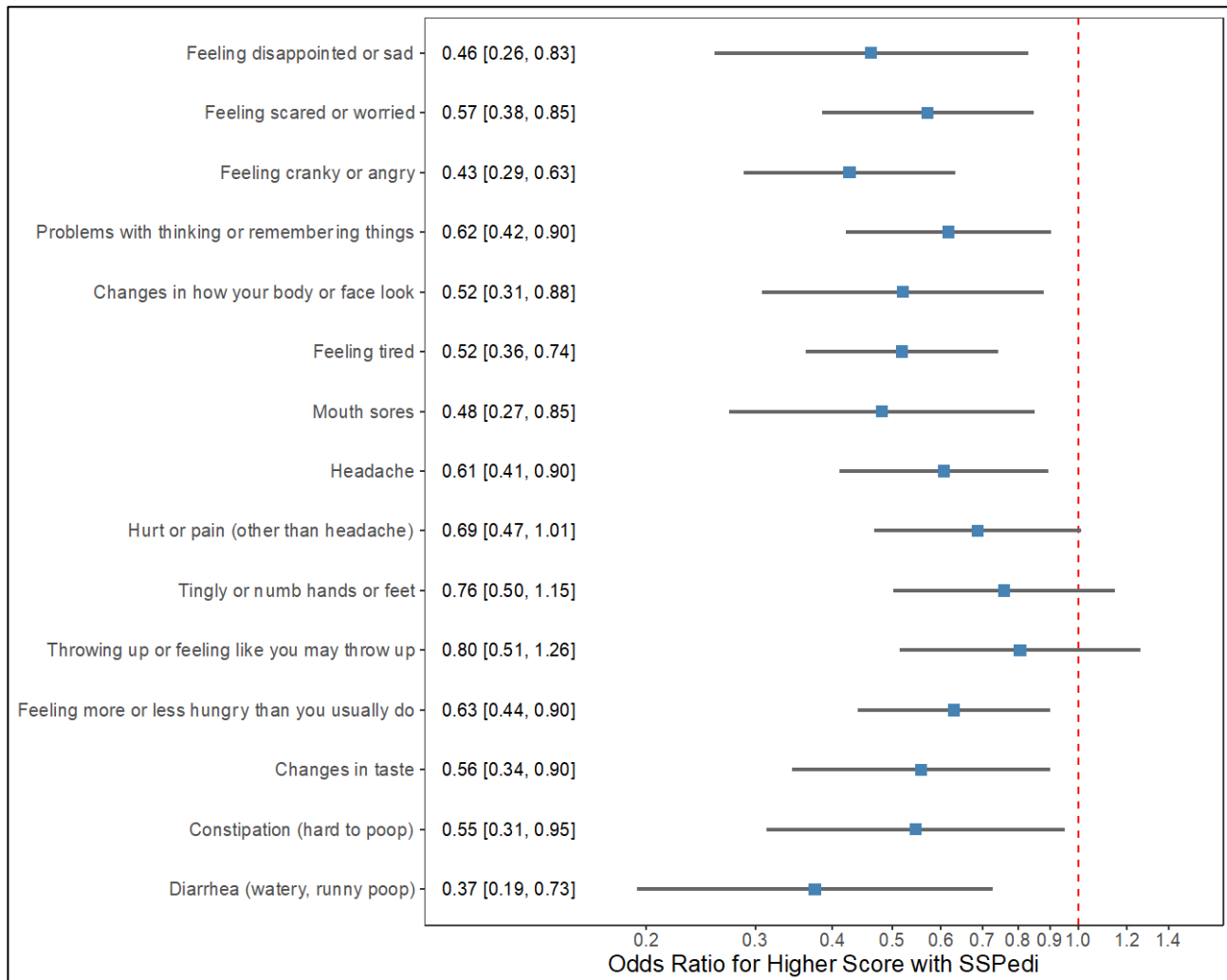
- b) Missing values are low: Within each site, missing values are sampled with replacement from the lowest 25% of observed values within the site. This presumes that all participants without week-8 SSPedi are doing as well as the least symptomatic participants at that site.
- c) Missing values are high: Within each site, missing values are sampled with replacement from the highest 25% of observed values at the site. This presumes that all participants without week-8 SSPedi are doing as poorly as the most symptomatic participants at that site.
- d) Best case for intervention: Within each control site, missing values are sampled with replacement from the highest 25% of observed values at the site. Within each intervention site, missing values are sampled with replacement from the lowest 25% of observed values at the site. This is an unrealistic scenario, as it presumes a completely different mechanism for non-response according to the site's treatment assignment.
- e) Worst case for intervention: Within each control site, missing values are sampled with replacement from the lowest 25% of observed values at the site. Within each intervention site, missing values are sampled with replacement from the highest 25% of observed values at the site. This is also an unrealistic scenario, as it presumes a completely different mechanism for non-response according to the site's treatment assignment.

	Mean Difference	Lower 95% CI	Upper 95% CI	P Value
Observed (complete case)	-3.8	-6.4	-1.2	.007
Multiple imputation using MICE	-3.5	-5.7	-1.4	.001
Impute from low quartile within site	-3.8	-5.7	-1.8	<.001
Impute from high quartile within site	-3.7	-6.9	-0.5	.03
Best case	-5.5	-8.5	-2.6	<.001
Worst case	-2.0	-4.3	0.3	.09

The interpretation of these analyses is that the primary results are robust to sensitivity analyses and different approaches to address missing data.

Abbreviation: CI – confidence interval; SSPedi – Symptom Screening in Pediatrics Tool; SPARK – Supportive care Prioritization, Assessment and Recommendations for Kids; PRO – patient-reported outcome; PROMIS - Patient Reported Outcomes Measurement Information System

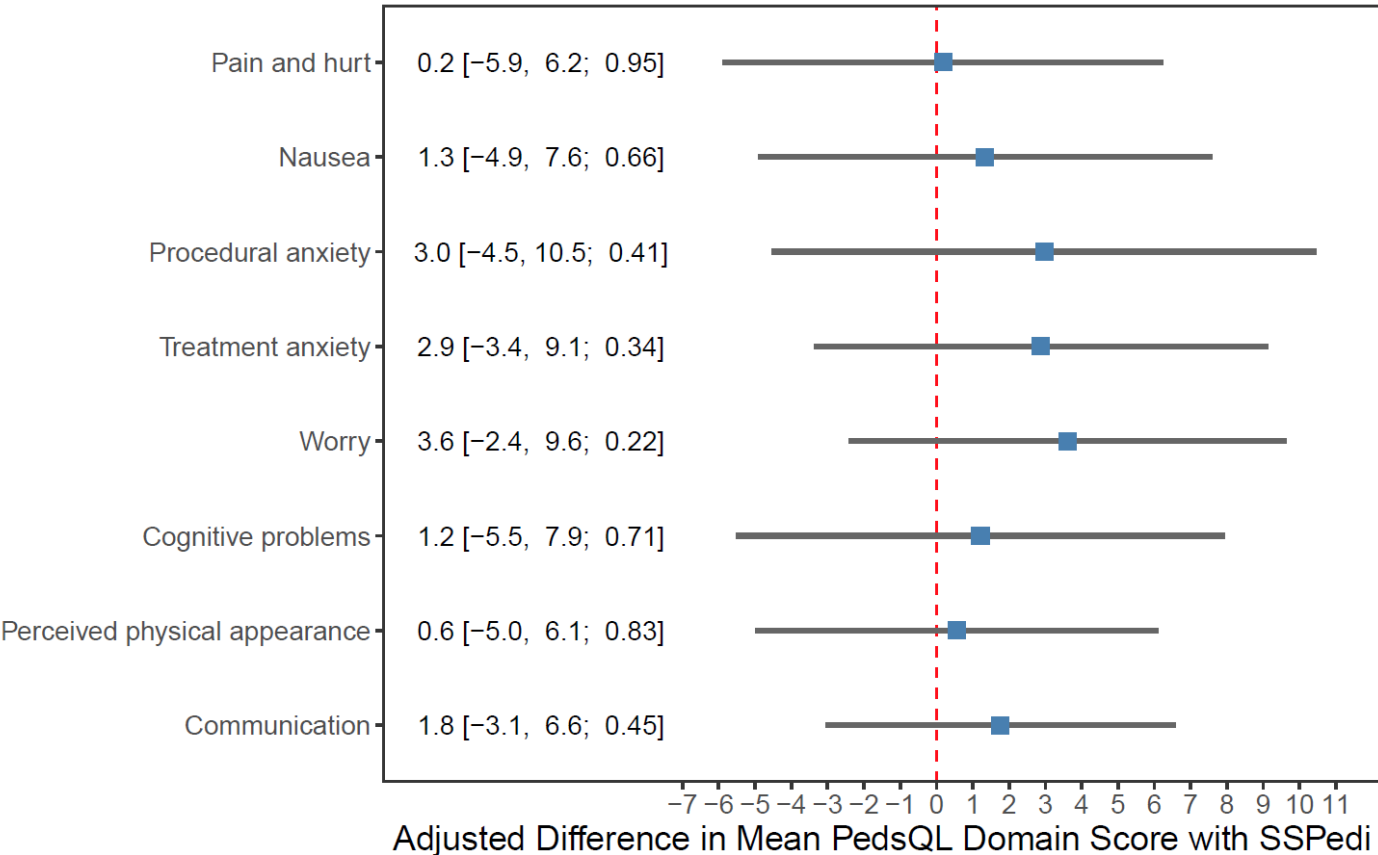
eFigure 2: Association between Symptom Screening and Higher (Worse) SSPedi Score for each Symptom included in SSPedi at Week 8



Legend: Odds ratio with 95% confidence interval from a proportional odds model adjusting for site and two stratification variables used for randomization. An odds ratio less than one suggests symptom screening is associated with a lower SSPedi scores for that symptom (less bothersome symptom).

Abbreviations: SSPedi – Symptom Screening in Pediatrics Tool

eFigure 3: Summary of Adjusted Estimates for Acute Cancer PedsQL Domain Scores at Week 8



Legend: Mean difference with 95% confidence interval from a mixed linear regression model with a random effect for site, fixed effects for treatment assignment, patient age and diagnosis group and two binary fixed effects at the cluster level for stratification variables. A value greater than 0 suggests symptom screening improves domain scores.

eText

eMethods

The primary analysis of week-8 total Symptom Screening in Pediatrics Tool (SSPedi) score adjusted for the stratification variables used for randomization. In addition, we adjusted for age and diagnosis group as they are associated with total SSPedi scores.⁴ With the primary analysis, we did not plan to adjust the model for baseline total SSPedi score because some of the care pathway interventions might affect symptoms prior to enrollment and the baseline score may already reflect some benefit of the cluster's treatment assignment. However, sensitivity analysis adjusted for baseline total SSPedi score. To account for potential differential enrollment of surgery-only patients (who might not be cared for by oncology teams) between intervention vs. control sites, we conducted a sensitivity analysis restricted to patients that received systemic chemotherapy.

For the analysis of documentation of symptoms, any intervention and symptom-specific intervention, each analysis fit a mixed effects logistic regression model to estimate the odds ratio for documentation, any intervention and symptom-specific intervention comparing intervention and control groups, adjusting for stratification factors. We anticipated there might be too few events to reliably fit this mixed model to all 15 symptoms. If this model could not be fit (when sigma cluster, or random effects variance could not be estimated), we fit a fixed effects logistic regression model only using treatment assignment.

The primary analysis of healthcare encounters used a mixed effects Poisson regression analysis with a random effect for site and fixed effects for the two stratification variables. Sensitivity analysis used negative binomial regression analysis, using both frequentist and Bayesian approaches. These adjusted for the same covariates as the Poisson regression analysis.

eResults

There was no difference in percentages declining participation between intervention (16.6%) and control (15.5%) sites ($P=.81$). However, those who agreed to participate were significantly older than those who declined participation (median age 14.8 vs. 13.2 years, $P=.001$). There were no significant differences by male sex (58.9% vs. 60.0%, $P=.94$) or cancer diagnosis (leukemia, lymphoma, solid tumor or brain tumor, $P=.13$).

In terms of those who were excluded due to physician preference, there was a significant difference by intervention (16.0%) and control (5.5%) sites ($P<.001$). Those participating compared to those excluded by their physician were not significantly different by age (median age 14.8 vs. 13.7 years, $P=.12$) or male sex (58.9% vs. 50.9%, $P=.33$). However, the distribution of cancer diagnoses differed ($P<.001$).

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